

EDUCATION.md — Learning Philosophy, Curriculum, Feedback & Quality Rules

Scope: everything about how the app teaches. Game economy is in GAME_DESIGN.md; per-child motivation details in CHARACTERS.md.

Learning philosophy

The AI acts like an **outstanding private tutor**. It must: build confidence; make learning enjoyable; encourage curiosity; explain the *why*, not just the answer; adapt to each child's learning style; celebrate effort, persistence, and improvement; challenge without overwhelming; encourage independence and critical thinking; and make mistakes feel like part of learning, never something to fear. Long-term goal: children who genuinely enjoy learning and succeed independently in school and beyond.

Grade & age model

- Age is auto-calculated from birthdays (CHARACTERS.md is the source of truth).
- Each child has a **current grade** that advances every school year and is manually adjustable by the parent. Content targets current grade → target grade. "Summer Bridge" (previous grade review → next grade prep) is an optional mode only.
- Current settings: Alex Grade 7→8, Katya Grade 4→5 (school year starting September 2026).

Curriculum mapping (Ontario)

All academic content follows the **Ontario curriculum** for the child's grade. French follows the **Ontario Core French** program (the children are in Core French, NOT French Immersion — do not use Immersion expectations).

Current math topic banks (question generators exist per topic, with fresh random numbers and themed stories):

- **Alex (Gr 7→8):** BEDMAS, fractions, decimals, percentages, integers, ratios & rates, algebra (solving for x), geometry, probability, money & finance, logic, word problems.
- **Katya (Gr 4→5):** multiplication, division, long division, fractions, decimals, money, time,

measurement, patterns, word problems — plus early literacy topics already present: reading, grammar, French.

Subject roadmap (feature-driven — no dates)

Math is the foundation and must reach the quality bar first. Then, in order: **1. Reading, 2. Grammar, 3. Writing, 4. Spelling, 5. Vocabulary, 6. French.** After those: Science, Social Studies, Geography, History, Financial Literacy, Coding & logical thinking, General knowledge, Study skills, Critical thinking, and eventually AI literacy and career exploration for older grades. **A new subject is added only when the previous one passes the Quality standards below.**

Language policy

- **English** = the app's only UI language and primary learning language. Never translate menus, navigation, parent controls, or system text.
- **French** = school subject only (Ontario Core French): vocabulary, reading, writing, grammar, listening, speaking, homework/test prep, uploaded teacher materials. Not an interface language.
- **Russian** = third-language subject (basic reading/writing, vocabulary, spelling, simple grammar, conversation, family phrases) **plus** warm cultural personalization (e.g., Grandma Olga encouraging Russian, bilingual challenges/stories). Secondary to school learning; never mandatory or pressure-based; never the app's main language.
- Each language has **separate progress tracking**.

Wrong-answer rule (mandatory for every learning flow)

The goal is learning, not punishment. A wrong answer must never feel like failure.

1. Allow up to **3 attempts** per question.
2. After each wrong attempt, give a **progressively more helpful hint** — never a bare "Incorrect."
3. On the 3rd wrong attempt: **explain the concept step by step and show the correct answer.**
4. Then generate **1–3 similar practice questions** to reinforce the concept; the child should answer correctly before moving on whenever reasonable.
5. Wrong answers never reduce XP, coins, or streaks. Reward persistence, improvement,

and effort instead.

6. The system identifies **recurring mistakes** and automatically schedules extra practice on those skills (error-pattern analysis + spaced repetition).

Status: the current app has hints and Brain Boost but does NOT fully implement this loop — closing that gap is a “Now” item in FUTURE_IDEAS.md.

Hint system: Brain Boost (progressive teaching ladder)

Each tap reveals one more teaching tier — never the answer to the live question. It teaches the METHOD using different numbers.

- **Daily/Guided missions (6 tiers):** 🧠 Brain Boost clue → ⚡ Winning Strategy → 🔍 Similar Example → 💡 Remember This (memory hook) → ⚠️ Common Mistake → 🎯 Bonus Challenge.
- **Battle (3 tiers, pace-friendly):** clue → strategy → common mistake.
- Content comes from the per-topic **Strategy Library**: each topic has a reusable strategy, a worked example (different numbers than any live question), a memory hook (visual or phrase), and the common student mistake — written to Ontario math teaching practice. Mental-math shortcuts live here (e.g., build percentages from 10% blocks; $9 \times 8 = 10 \times 8 - 8$; count UP for change; BEDMAS order; sign rules via “same signs happy, different signs grumpy”).
- Viewing/using strategies is tracked and unlocks strategy badges; there is a browsable Strategy Library screen.
- New shortcuts, hooks, and strategies belong in the Strategy Library entry for their topic — not scattered in question text.

Feedback & motivation rules

- Praise effort, persistence, careful thinking, responsibility, and improvement — not innate smartness, not speed alone.
- Separate identity from performance; never shame, humiliate, or use fear.
- Never compare the children against each other; sibling challenges stay friendly and optional.
- Explain the purpose of tasks (especially for Katya — see CHARACTERS.md).
- Encourage slowing down, checking answers, and quality (especially for Alex).

- School marks are one signal, never the only measure of success.

Daily expectations & session limits

- **Minimum daily goal:** ~15–20 minutes or one Daily Mission (~20 questions, difficulty-dependent).
- The AI adapts question count to subject, grade, difficulty, the child's attention and performance, and time available.
- **Maximum recommended session:** ~30–40 minutes per subject, then encourage a break or activity switch. If the child is highly engaged, offer optional **Bonus Missions** rather than forcing a stop.

Quality standards (gate for calling a subject "done")

A subject is complete only when Yulia would confidently let the children use it daily **without supervision**. Checklist per subject:

- Full Ontario curriculum coverage for the appropriate grades
- Complete topic progression, beginner → advanced
- Multiple question types (not just multiple choice)
- Personalized hints and explanations
- Adaptive difficulty
- Mastery tracking by skill
- Spaced repetition
- Error-pattern analysis
- Real-world examples
- Fun game mechanics and engaging stories
- Excellent visual design
- Fast performance; no obvious bugs
- Parent dashboard integration
- Support for uploaded school homework and worksheets
- AI tutoring that teaches concepts instead of giving answers

